From what we've built to what the chief needs.

Firewatch: the current build compared with the proposed design

Team Watch Dogs. Firewatch 2026.

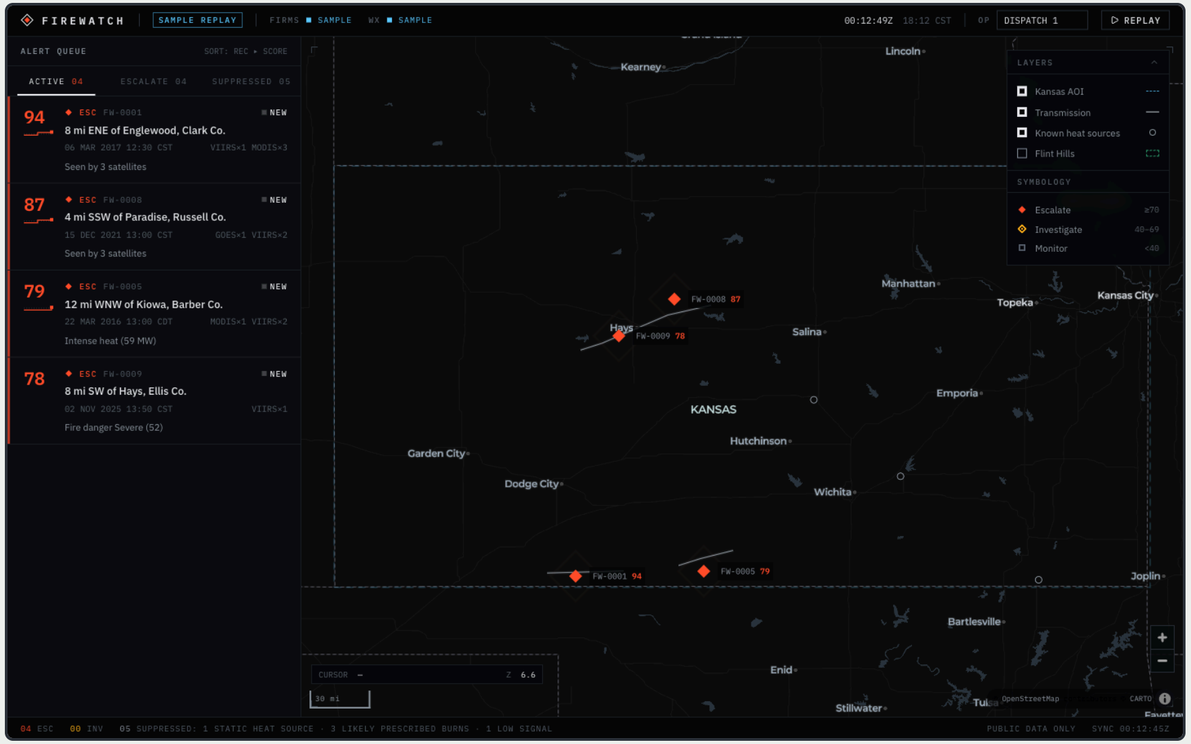
Two things stay, five change and eight are new.

What it does	Current build	Proposed	Change
Ranked alert queue, score and reasons	Yes	Yes, plus burn-log reasons	Keep
Spread forecast model	Yes, 3D preview	Same model, shown as who is in the path	Change
Satellite heat from FIRMS	Yes	Yes	Keep
Satellite heat from GOES-19	No	Read directly every 5 to 10 minutes	New
Telling burns from fires	Rules guess prescribed burns	Burns logged by dispatch, rules as backup	Change
Data age per source	Two sources, no age in sample mode	Every source shows its age	Change
Ack, escalate, dismiss with a log	Yes. Reopen only, no undo	Same, with undo	Change
Alert screen	Desktop drawer	Daylight phone screen, one action	Change
Phone view for the chief	No	Yes	New
Text and pager alerts	No	Push, text, radio pager	New
Power-line fault input	No	From the co-op	New
Public reports	No	By text, held until a bar	New
Offline district map	No	Yes	New
Standard alert export (CAP)	No	Yes	New
Timeline of today vs Firewatch	No	Yes, from the simulation	New

Source: NOAA GOES-19 fire product and NASA FIRMS (Earthdata, ARSET 2025), as summarised in our research notes

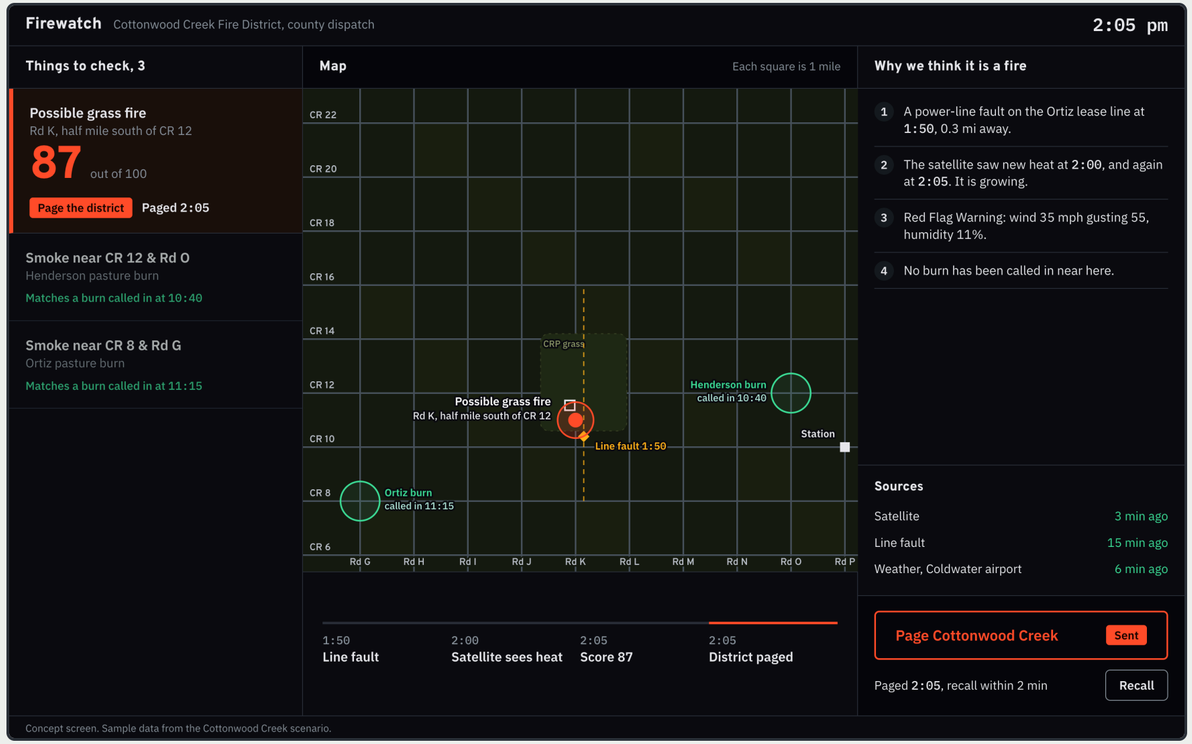
The dispatcher keeps the ranked queue and gains the reason, the data age and a page action.

Current build



Stays: ranked queue, 0 to 100 score, reasons.

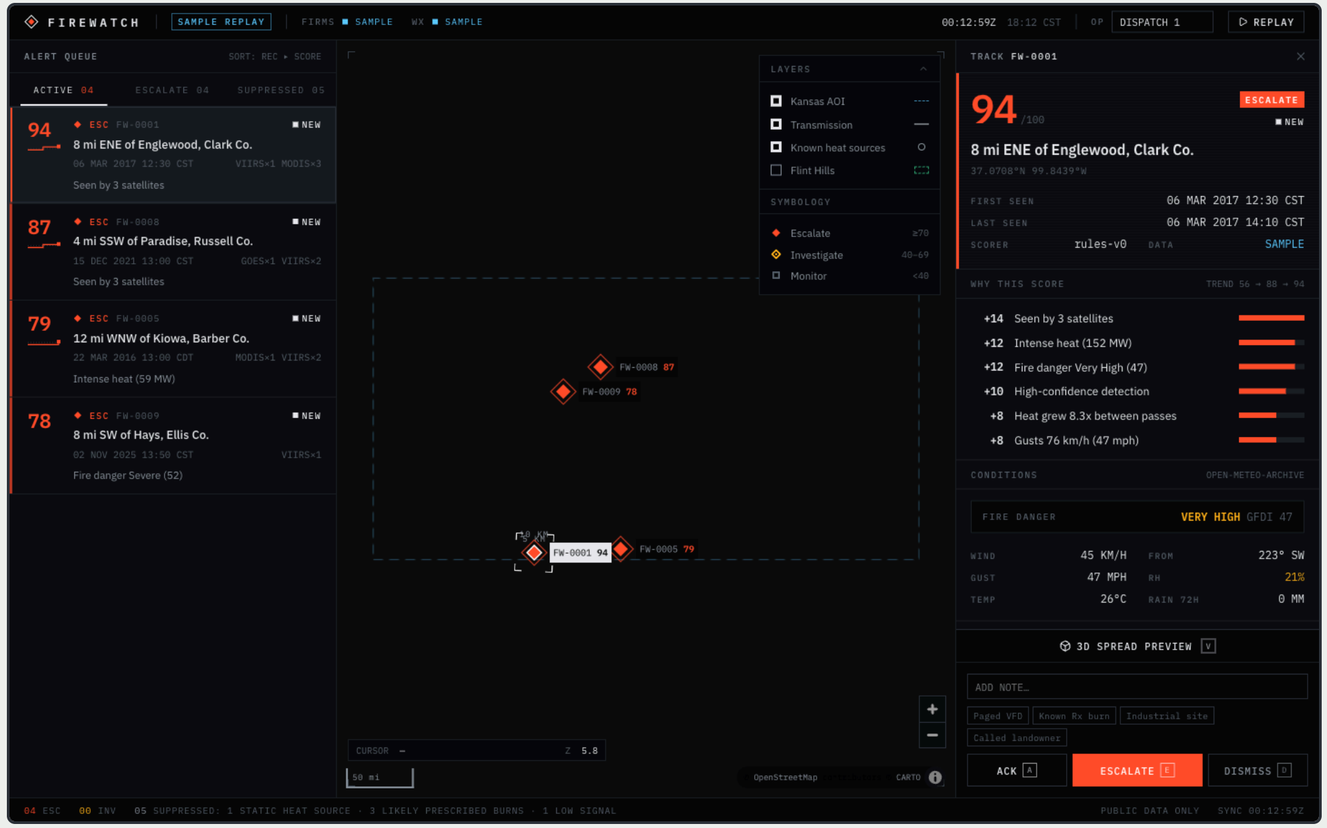
Proposed



Changes: reasons name the burn log, each source shows its age, and a "Page the district" action is added.

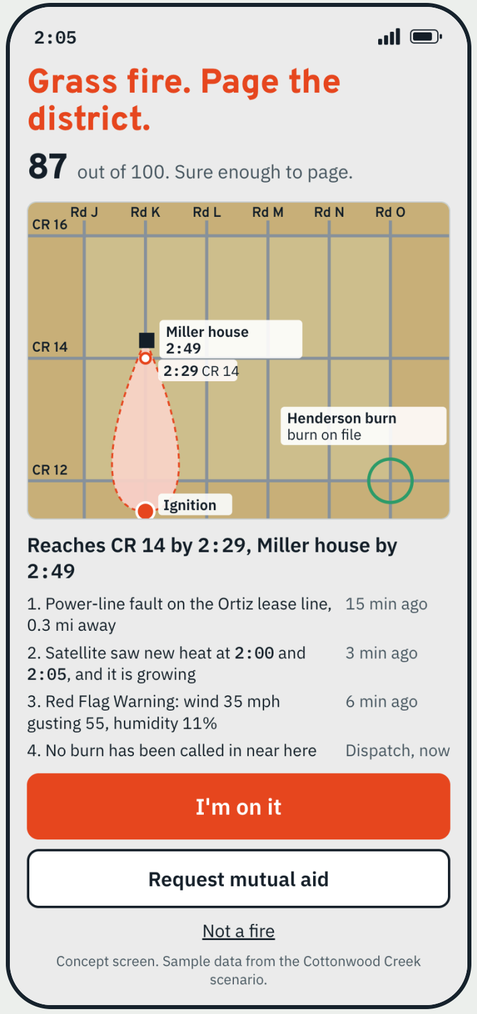
A desktop drawer becomes a daylight phone screen with one action.

Current build



The drawer has Ack, Escalate and Dismiss, a log and a table of detections, built for a seated operator.

Proposed



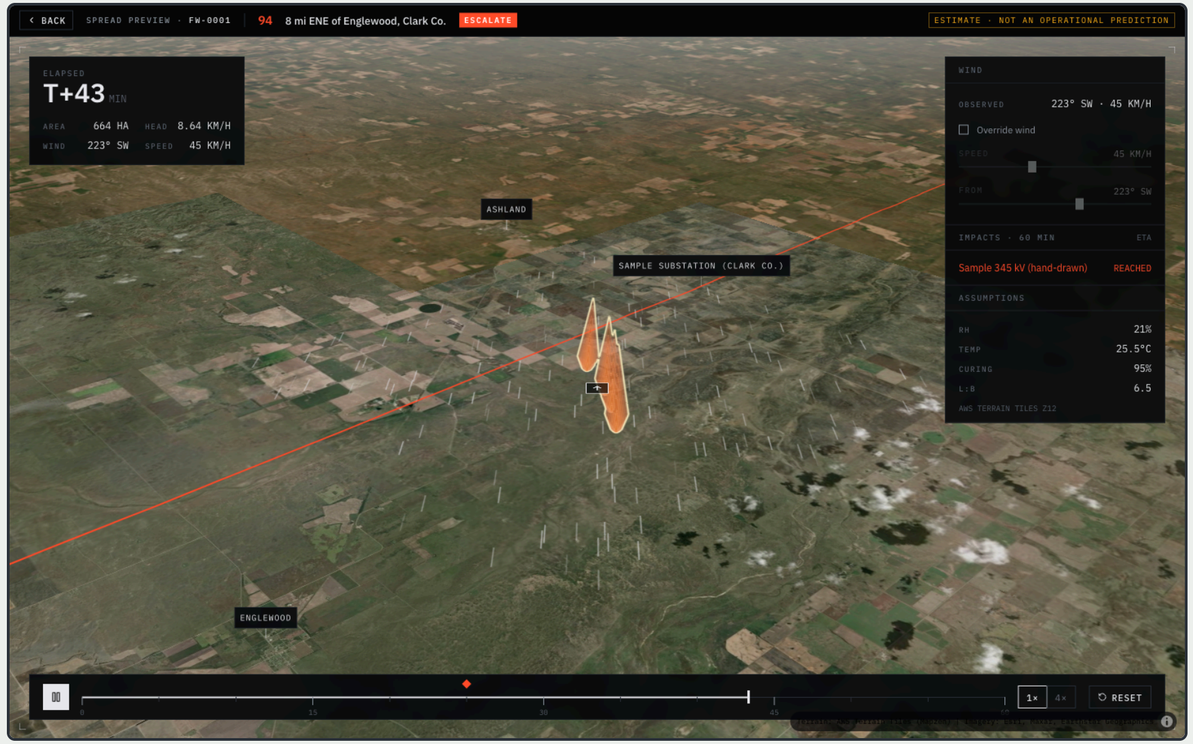
Big buttons, plain words, readable in direct sun.

Current build → proposed

Alert detail

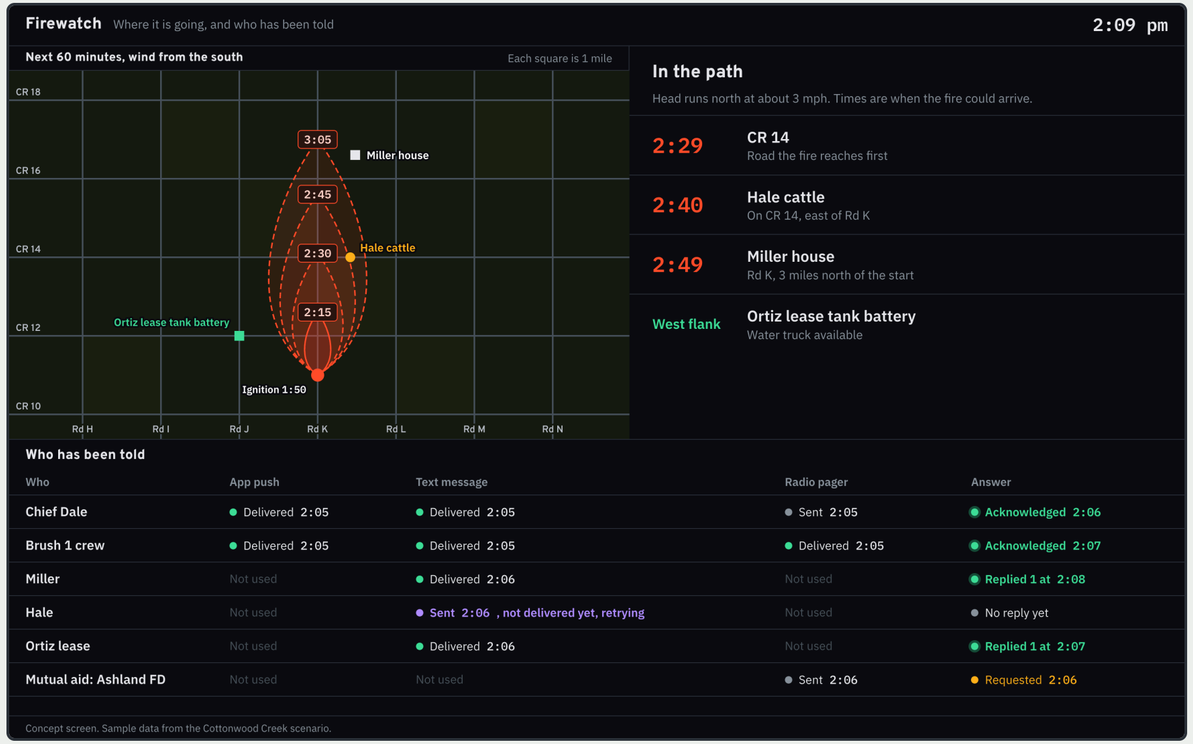
The 3D wind preview becomes "who is in the path" with delivery status. The spread model is reused.

Current build



3D terrain with a 60-minute spread and a wind override. Impacts list assets such as power lines and substations.

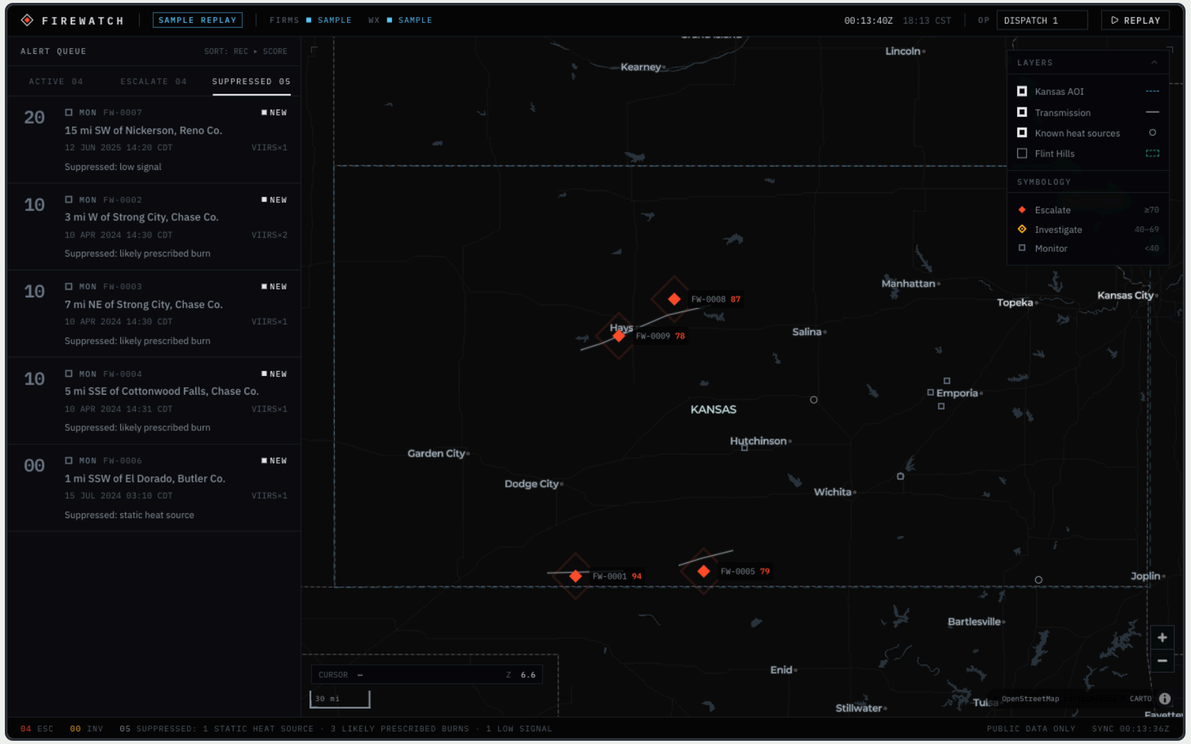
Proposed



The same forecast, drawn on the section grid. It lists houses and roads with arrival times, and shows who was texted.

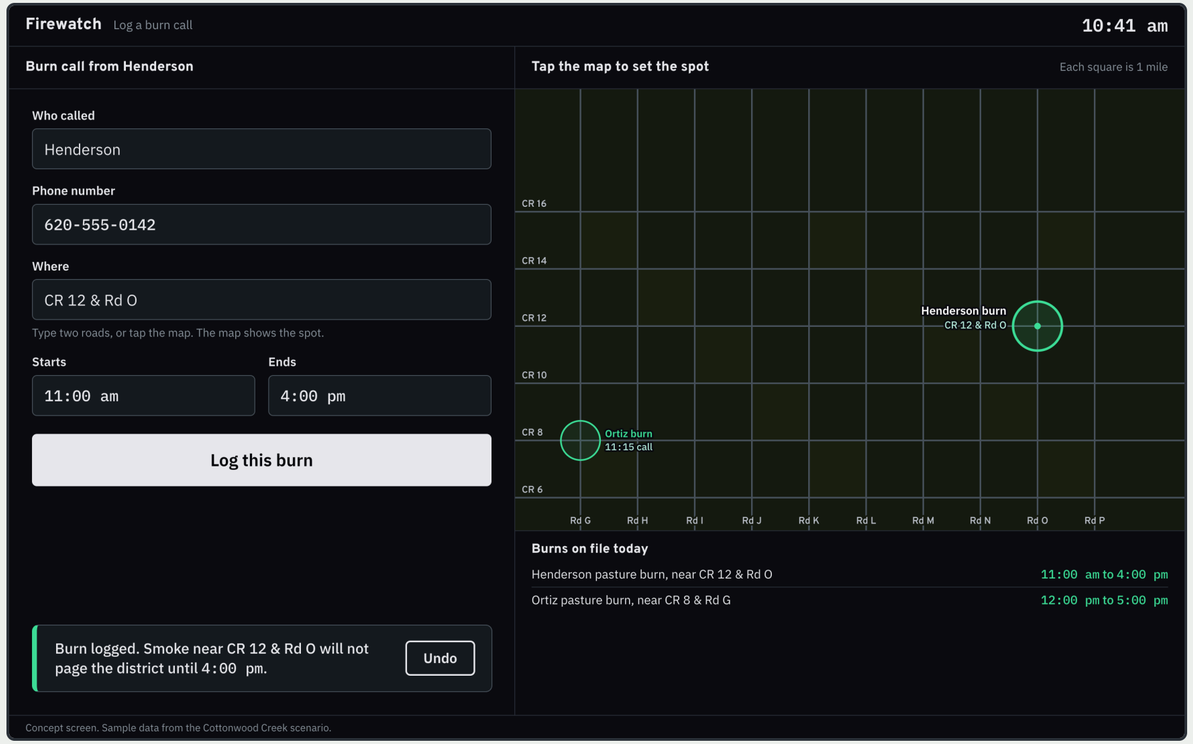
Rule-based burn suppression becomes burns logged by dispatch, with the rules kept as backup.

Current build



The app sorts alerts into a Suppressed tab with a reason: likely prescribed burn (Flint Hills layer), known industrial heat source, or low signal. It guesses.

Proposed



Dispatch logs each burn with place and time. A hotspot near one scores down and says "matches a burn called in at 10:40". The old rules stay as a second check.

Eight pieces do not exist yet, and the chief needs most of them.

Phone view

A screen the chief can use in the truck

To build

Text and pager delivery

Push, text and county radio pager

To build

Line-fault input

Events from the electric co-op

To build

Public reports

Texts that join the right fire

To build

Offline map

District map that loads with no signal

To build

Standard alert export

CAP file for dispatch and Everbridge

To build

Timeline simulation

Today vs Firewatch, minute by minute

To build

GOES-19 direct ingest

Satellite heat from NOAA open data

To build

Current build → proposed

Not built yet

The API gains five ways in and out, and the existing eleven stay.

Today (under /api)

Alerts: list, detail, actions
Spread simulation, with wind override
Map layers
Health
Manual satellite poll (live mode)
Sample replay, load and fixtures

No endpoints for notifications, reports, logs export, CAP or timelines.

To add

POST /api/signals/line-fault
a fault from the co-op
POST /api/signals/report
a public text or mesh report
POST /api/burns
a burn called in to dispatch
GET /api/alerts/{id}.cap.xml
standard alert export
Alert fan-out
push, email-to-page and text (stub)

We build in order, cut from the bottom, and protect the demo.

Order of work

1. Cottonwood scenario and burn log
2. Line-fault and GOES input
3. Phone alert, push and pager
4. Who is in the path
5. CAP export and offline map

Cut lines

If GOES ingest fails, replay recorded detections and call live ingest next.

Push plus the path list is enough. CAP and a better spread model are extras.

Deck and video are never cut.

Risks of the change

Scope: the Sunday `10:00` code freeze is close.

Demo stability: every addition can break the replay.

Sample mode must keep working, so new inputs need fixtures.